Utkarsh Sinha

Data Scientist & Generative AI Engineer

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PROFESSIONAL SUMMARY

Data Scientist & Generative AI Engineer with 3 years of experience building production-grade AI systems across NLP, financial modeling, and LLM-based architectures. Specialized in designing scalable MLOps pipelines, fine-tuning large language models (32B–70B), and deploying RAG-powered applications on AWS. Strong expertise in Transformers, time-series forecasting, vector search systems, and autonomous AI agents.

SKILLS

- **Programming & AI:** Python (Automation, DSA) | Machine Learning | Deep Learning (ANN, PyTorch)
- **Natural Language Processing:** RNN | LSTM | Encoder-Decoder | Attention | Transformer | BERT | GPT
- **LLM Frameworks:** LangChain | LlamaIndex | AWS Bedrock
- **Relational & NoSQL Databases:** MySQL | MongoDB
- **MLOps Tools:** Git | Docker | DVC | MLflow | GitHub Actions | Langsmith
- **Computer Vision:** CNN | Image Processing | Object Classification
- **Generative AI:** OpenAI | LLaMA | Gemini | Hugging Face | RAG | Fine-tuning | AI Agents | Embeddings
- **Vector Databases:** FAISS | ChromaDB | Pinecone | LanceDB
- **REST APIs:** FastAPI | Flask
- **Cloud Platforms:** AWS

PROFESSIONAL EXPERIENCE

Data Scientist & GenAI Engineer

LawSikho & Skill Arbitrage — *1st May 2023 – 5th November 2026*

- Led a team of 2+ AI engineers to architect and deploy production-ready ML and GenAI systems across education and HR tech domains.
- Designed end-to-end MLOps pipelines reducing model deployment time by ~40% using MLflow, DVC, Docker, and GitHub Actions.
- Delivered AI consulting projects impacting 1,000+ learners and internal stakeholders.
- Conducted 1,000+ live training sessions on Data Science, MLOps, and Generative AI with consistent 4.5+/5 feedback ratings.
- Standardized reusable ML templates, accelerating future project delivery by ~30%.

ML & AI Engineer

SmartXAlgo — *29th April 2026 – Present*

- Architected an AI-driven stock trading automation platform integrating time-series forecasting, NLP sentiment modeling, and LLM-based reasoning, achieving ~70% directional prediction accuracy in back testing.
- Built a modular multi-model decision engine improving signal validation robustness through ensemble approval workflows and 3 other AI Agent workflow.
- Automated CI/CD-based model deployment pipelines ensuring reproducible financial model releases.

PROJECTS

Stock Market AutoTrader – Multi-Model AI Trading Architecture

SmartXAlgo

- Developed a Transformer-based multi-horizon forecasting model trained on 5+ years of minute-level OHLCV data (2,000+ equities).
- Boosted directional prediction consistency by integrating volatility, momentum, and regime-aware indicators up to ~80%.
- Fine-tuned BERT for financial sentiment classification achieving higher contextual accuracy on news-driven events accuracy upto ~93% no over-fitting.
- Fine-tuned 32B Qwen/DeepSeek models using LoRA (Unsloth), reducing GPU memory requirements by ~60%.
- Created RAG-based financial reasoning agent integrating live market retrieval (Tavily API).
- Designed multi-layer validation engine combining forecast + sentiment + fundamentals + LLM reasoning before execution.
- Deployed modular auto-trade confirmation workflow using agent-based architecture.

TalentLens

LawSikho & Skill Arbitrage

- Engineered real-time RAG-powered CV matching engine processing structured and unstructured resume data.
- Improved retrieval precision by combining embedding similarity with TF-IDF hybrid scoring improved retrieval precision by 95%.
- Processed 1,000+ CVs using unstructured document parsing pipelines.
- Reduced candidate-role matching time by ~70% compared to manual screening.
- Incorporated semantic reranking using DeepSeek R1 Distill LLaMA-70B via Groq.
- Deployed on AWS EC2 with Dockerized architecture and automated CI/CD pipeline.

MLOPs & Pipeline Standardization
LawSikho & Skill Arbitrage

- Orchestrated modular ML pipelines enabling reproducible experimentation and version control across multiple projects.
- Integrated DVC with AWS S3 for dataset versioning across large-scale model training workflows.
- Implemented real-time monitoring using Prometheus & Grafana, reduced model drift detection latency by 50%.
- Automated Docker-based deployments through GitHub Actions and AWS ECR, reducing deployment time by 60% and enabling daily production releases.
- Project GitHub Links:
    - Capstone-Project: https://github.com/utkodex/Capstone-Project
    - InsureFlow-MLOps: https://github.com/utkodex/InsureFlow-MLOps

AWARDS

The Innovation Mavrick - Novermber 2023
LawSikho & Skill Arbitrage

EDUCATION

BBA LLB
Guru Gobind Singh Indraprastha University
Cumulative Percentage: 66.0%